

A machine-checked, uniform-in-order Abel formula for the Casoratian, in Lean 4/Mathlib

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Abstract

For k sequences $s^{(0)}, \dots, s^{(k-1)}: \mathbb{N} \rightarrow R$ over a commutative ring R , all solving the order- k homogeneous linear recurrence $s(n+k) = \sum_{j=0}^{k-1} c_j(n) s(n+j)$, the *Casoratian* (discrete Wronskian) $C(n) = \det [s^{(t)}(n+i)]_{0 \leq i, t < k}$ obeys the first-order law $C(n+1) = (-1)^{k-1} c_0(n) C(n)$, hence the closed product form $C(n) = (\prod_{l < n} (-1)^{k-1} c_0(l)) C(0)$. Only the *lowest* coefficient c_0 survives; $c_1, \dots, c_{k-1}$ cancel. This is *Abel's formula* for the Casoratian — classical, going back to Casorati, the discrete analogue of Liouville's formula for the Wronskian. Our contribution is not the identity but its treatment: we give a single proof valid *uniformly in the order k* and formalize it in Lean 4/Mathlib to a clean axiom cone (a subset of `{propext, Classical.choice, Quot.sound}`, no `sorry`). The formal proof rewrites the last row of $C(n+1)$, via the recurrence, into a linear combination of the *cyclically rotated* rows of $C(n)$; `Matrix.det.updateRow_sum` collapses the combination to the single coefficient $c_0(n)$ and `sign_finRotate` supplies the cyclic sign $(-1)^{k-1}$. We also formalize the constant-coefficient power form $C(n) = ((-1)^{k-1} c_0)^n C(0)$, an integral-domain non-vanishing (linear-independence) certificate, and the order-2 and order-3 instances proved independently by hand as faithfulness witnesses (order 2 reproduces the corpus sign $-c_0$, with Euler's $(-1)^n$ convergent Wronskian as the case $c_0 \equiv 1$; order 3 gives $+c_0$). The result opens the order- ≥ 3 direction in a formalized polynomial-continued-fraction corpus previously confined to the order-2 (three-term) case — the algebraic backbone of simultaneous / Hermite–Padé approximation and of Apéry-type irrationality certificates. An independent computer-algebra gate confirms the identity symbolically at $k = 3$, by exact-rational checks for $k = 2, \dots, 7$, and on named instances.

2020 MSC: 39A06 (primary), 68V20, 11J72, 15A15. **Keywords:** Casoratian, discrete Wronskian, Abel's formula, linear difference equation, higher-order recurrence, Hermite–Padé approximation, Apéry-type irrationality, Lean 4, Mathlib, formal verification, polynomial continued fractions.

1 Introduction

The convergents of a polynomial continued fraction $K_{n \geq 1} a_n/b_n$ are governed by a three-term recurrence $q_n = b_n q_{n-1} + a_n q_{n-2}$, and the basic structural invariant is the order-2 Casoratian (discrete Wronskian) of two solutions: it flips sign and scales by the lower coefficient at each step, so the convergent Wronskian $p_{n+1} q_n - p_n q_{n+1} = (-1)^n$ is constant up to sign. This is the certificate behind Euler's irrationality criterion, and across the deposited corpus the Casoratian appears only in this order-2, three-term form ([12, 11]; formalized as `casoratian_eqG` in the companion `pcf-delta` project [13]).

The order- ≥ 3 direction. The natural next object is the Casoratian of $k \geq 3$ solutions of an order- k recurrence

$$s(n+k) = \sum_{j=0}^{k-1} c_j(n) s(n+j), \quad c_0, \dots, c_{k-1}: \mathbb{N} \rightarrow R, \quad (1)$$

the $k \times k$ determinant

$$C(n) = \det [s^{(t)}(n+i)]_{0 \leq i, t < k}. \quad (2)$$

This is the algebraic backbone of *simultaneous* (Hermite–Padé) approximation, the engine of Apéry-type irrationality proofs ([4]): there one builds several sequences satisfying one common higher-order recurrence and certifies their linear independence through the non-vanishing of exactly this determinant. The first-order law that C obeys is classical — the discrete analogue of Abel’s identity for the Wronskian of a linear ODE, going back to Casorati; see, e.g., [1, Ch. 2] and [2]. It states that C is reproduced at each step by a single scalar built only from the *lowest* coefficient c_0 .

Contribution. The identity itself is not new. What we provide is:

- (i) a single proof valid *uniformly in the order k* (Theorem 1), and its formalization in Lean 4 with Mathlib to a clean axiom cone (Section 5), so the result is `PROVEN` in the strict sense of this program: `#print axioms` reports a subset of `{propext, Classical.choice, Quot.sound}` with no `sorryAx` and no `sorry/admit/native_decide` in source;
- (ii) the closed product form, the constant-coefficient power form, and an integral-domain non-vanishing (linear-independence) certificate, all formalized (Section 4);
- (iii) the order-2 and order-3 instances proved *independently* by a direct cofactor computation, kept as faithfulness witnesses against the general proof (Proposition 1); and
- (iv) an independent computer-algebra gate (Section 6) that re-derives the law symbolically and on named instances, in the program’s “gate-before-Lean” discipline.

Concretely, this opens the order- ≥ 3 space inside a formalized PCF corpus that had treated only order 2, and supplies a reusable, machine-checked independence certificate for future Hermite–Padé constructions.

2 The Casoratian of a linear recurrence

Throughout, R is a commutative ring and $k \geq 1$ a fixed order. We write a solution of (1) simply as a sequence $s: \mathbb{N} \rightarrow R$ satisfying that identity for all n ; no invertibility of the coefficients is assumed.

Definition 1 (Casoratian). *For sequences $s^{(0)}, \dots, s^{(k-1)}: \mathbb{N} \rightarrow R$, the Casoratian at n is the determinant (2) of the $k \times k$ matrix whose (i, t) entry is the i -shift of the t -th sequence, $s^{(t)}(n+i)$.*

Because $\det$ is invariant under transposition, it is immaterial whether one indexes rows by shifts and columns by solutions or vice versa; we fix rows = shifts. The two “ends” of $C(n+1)$ are what drive the law:

- **Top $k-1$ rows.** For $0 \leq i \leq k-2$, row i of $C(n+1)$ is the shift vector $(s^{(t)}(n+1+i))_t$, which is precisely row $i+1$ of $C(n)$. So the top $k-1$ rows of $C(n+1)$ are the bottom $k-1$ rows of $C(n)$, in order.
- **Bottom row.** Row $k-1$ of $C(n+1)$ is the shift vector at $n+k$, $(s^{(t)}(n+k))_t$, which by the recurrence (1) equals $\sum_{j=0}^{k-1} c_j(n) (s^{(t)}(n+j))_t$, a linear combination of *all* k rows of $C(n)$.

3 The general-order Abel formula for the Casoratian

Theorem 1 (Abel’s formula for the Casoratian, uniform in k). *Let $s^{(0)}, \dots, s^{(k-1)}: \mathbb{N} \rightarrow R$ all satisfy the order- k recurrence (1). Then the Casoratian (2) satisfies the first-order law*

$$C(n+1) = (-1)^{k-1} c_0(n) C(n) \quad (n \in \mathbb{N}), \quad (3)$$

and consequently the closed product form

$$C(n) = \left(\prod_{l=0}^{n-1} (-1)^{k-1} c_0(l) \right) C(0) = (-1)^{(k-1)n} \left(\prod_{l=0}^{n-1} c_0(l) \right) C(0). \quad (4)$$

Only the lowest coefficient c_0 enters; $c_1, \dots, c_{k-1}$ cancel.

Proof. Let ρ be the cyclic permutation of $\{0, 1, \dots, k-1\}$ that moves every index up by one and wraps the top to the bottom, $\rho(i) = i+1$ for $i < k-1$ and $\rho(k-1) = 0$; it is a single k -cycle, so $\text{sgn } \rho = (-1)^{k-1}$. Let M be $C(n)$ with its rows permuted by ρ , i.e. row i of M is row $\rho(i)$ of $C(n)$. By the observation in Section 2, the top $k-1$ rows of M (namely rows $1, \dots, k-1$ of $C(n)$) are exactly the top $k-1$ rows of $C(n+1)$, while the bottom row of M is row 0 of $C(n)$.

Replacing the bottom row of M by the linear combination $\sum_{j=0}^{k-1} c_j(n)$ (row j of $C(n)$) produces exactly $C(n+1)$, because that combination is the bottom row of $C(n+1)$ and the top rows already agree. Now take determinants. Multilinearity in the replaced row — the content of `Matrix.det_updateRow_sum` — gives

$$\det C(n+1) = \sum_{j=0}^{k-1} c_j(n) \det(M \text{ with bottom row } = \text{row } j \text{ of } C(n)).$$

Every summand with $j \geq 1$ places into the bottom position a row of $C(n)$ that already occurs among the top $k-1$ rows of M , so that determinant vanishes. Only $j = 0$ survives, and there the bottom row is the original row 0 of $C(n)$, so the summand is $c_0(n) \det M$. Finally $\det M = \text{sgn } \rho \cdot \det C(n) = (-1)^{k-1} \det C(n)$. Combining gives (3). A trivial induction on n , using $C(0)$ as the base case and (3) at the inductive step, yields the product form (4). $\square$

Remark 1 (What the formalization does, verbatim). *The Lean development (Section 5) realizes this proof literally. It builds M as `(casoMats n).submatrix (finRotate (k+1)) id` (Mathlib’s `finRotate` is the cycle ρ), expresses $C(n+1)$ as that submatrix with its last row updated to the recurrence combination, then rewrites with `Matrix.det_updateRow_sum` (the multilinear collapse to the surviving coefficient) and `Matrix.det_permute` together with `sign_finRotate` (the cyclic sign). The “vanishing for $j \geq 1$ ” step is automatic: it is exactly the statement that `det_updateRow_sum` isolates the coefficient indexed by the updated row. No analytic input is used; the argument is finitary algebra over an arbitrary `CommRing`.*

4 Specializations and consequences

Corollary 1 (Constant lowest coefficient). *If $c_0(l) = a$ for all l , then $C(n) = ((-1)^{k-1} a)^n C(0)$. In particular, if a is a unit (or 1), the Casoratian never vanishes once $C(0) \neq 0$.*

Corollary 2 (Non-vanishing / linear-independence certificate). *Suppose R is an integral domain. If $C(0) \neq 0$ and $c_0(l) \neq 0$ for every $l < n$, then $C(n) \neq 0$. Hence if $C(0) \neq 0$ and c_0 is nowhere zero, the k solutions stay linearly independent at every window: $C(n) \neq 0$ for all n .*

Proof. Immediate from (4): a product of nonzero elements of a domain is nonzero, and $(-1)^{k-1}$ is a unit. $\square$

Corollary 2 is the order- k analogue of the order-2 statement $p_{n+1}q_n - p_nq_{n+1} = (-1)^n \neq 0$ that underlies Euler’s criterion: it is precisely the determinantal independence certificate one needs for Hermite–Padé systems.

Proposition 1 (Low-order witnesses). *The cases $k = 2, 3$ hold and are proved independently by direct cofactor expansion, without the permutation argument:*

$$k = 2: \quad C(n+1) = -c_0(n)C(n); \quad k = 3: \quad C(n+1) = +c_0(n)C(n).$$

The order-2 sign $-c_0$ is the corpus sign; its $c_0 \equiv 1$ instance is the constant convergent Wronskian $p_{n+1}q_n - p_nq_{n+1} = (-1)^n$.

These hand proofs (`caso2_step`, `caso3_step`; the 3×3 cofactor expansion is closed by `ring`) are retained as *faithfulness witnesses*: they agree with Theorem 1 at $k = 2, 3$, guarding against a vacuous or mis-signed general statement.

5 The machine-checked formalization

The development is in Lean 4 with Mathlib (toolchain pinned `leanprover/lean4:v4.30.0`, Mathlib `rev=v4.30.0`). It is built and axiom-checked inside the companion `pcf-delta` `PcfContinuant` project [13], whose `Check.lean` gate runs `#print axioms` on every declaration; the two source files `GeneralCaso.lean` (the uniform-in- k core, namespace `PcfGeneralCaso`) and `HigherCaso.lean` (the by-hand low-order witnesses, namespace `PcfHigherCaso`) are mirrored in this catalogue under `verify/`.

Remark 2 (Index convention). *The formalization parametrizes the order as $k+1$ via `Fin (k+1)` (so the recurrence is $s(n+(k+1)) = \sum_{j \leq k} c_j(n)s(n+j)$ and the matrix is $(k+1) \times (k+1)$); its sign therefore reads $(-1)^k$. This is the prose order- k law (3) under $k \mapsto k+1$: an index shift only, with $(-1)^{(k+1)-1} = (-1)^k$.*

Theorem 2 (Formalized core; PROVEN). *The following nine declarations are machine-checked with clean axiom cones — each a subset of $\{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}$, no `sorryAx`, sources free of `sorry/admit/native_decide`, `lake build` exit 0.*

- Uniform in k (`PcfGeneralCaso`, `GeneralCaso.lean`): `casoMat_det_step` is the step law (3) $C(n+1) = (-1)^k c_0(n)C(n)$ (order $k+1$; Remark 2); `casoMat_det_eq` is the closed product form (4); `casoMat_det_eq_const` is the constant-coefficient power form (Corollary 1); `casoMat_det_ne_zero_of_init` is the integral-domain non-vanishing certificate (Corollary 2).
- Low-order witnesses (`PcfHigherCaso`, `HigherCaso.lean`): `caso2_step` (the order-2 sign $-c_0$), `caso3_step` (the order-3 step $+c_0$), `caso3_eq` (its product form), `caso3_eq_const` (its constant-coefficient case), and `caso3_ne_zero_of_init` (its domain non-vanishing certificate) — Proposition 1.

Of the nine, exactly two close with the minimal cone $\{\text{propext}, \text{Quot.sound}\}$ — the by-hand witnesses `caso2_step` and `caso3_step`; the remaining seven additionally use `Classical.choice`. This `Classical.choice` usage is inherited from Mathlib’s general determinant infrastructure (`Matrix.det` and the underlying permutation machinery), not a sign that the uniform-in- k proof is in any meaningful sense less constructive than the by-hand cases; we report the cones factually rather than rank them.

This follows the program’s *unconditional-core* pattern: there is no labelled analytic hypothesis here at all, since the statement is pure finitary algebra over a `CommRing` (the domain results add only `IsDomainR`). The Casoratian is defined as a genuine `Matrix` determinant, so the result is the honest determinantal identity, not a bespoke $k = 2, 3$ polynomial expression. Only the order-2 *step* sign is recorded here, as `caso2_step`: the full order-2 closed-form suite already exists in `pcf-delta` as `casoratian_eqG` [13] and is not duplicated, whereas order 3 carries the full suite as the smallest genuinely higher-order witness.

6 Independent symbolic verification

In keeping with the catalogue’s gate-before-Lean discipline, a `sympy` harness (`harness_caso_k.py`) re-derives the law independently of the Lean proof:

- a fully *symbolic* proof at $k = 3$: with generic symbolic data and generic c_0, c_1, c_2 , $\det C(1) - c_0 \det C(0)$ expands to 0 and $\det C(1)$ is free of c_1, c_2 (only c_0 enters);
- *exact-rational* checks of both the step law (3) and the product form (4) for $k = 2, \dots, 7$, over integer-coefficient recurrences and initial data drawn from a *fixed deterministic seed* (`random.Random(7k+101t+5)` for trial t), so the runs are exactly reproducible, confirming the alternating sign $(-1)^{k-1}$; and
- named instances: the Tribonacci recurrence $s(n+3) = s(n+2) + s(n+1) + s(n)$ ($c_0 \equiv 1$) has a *constant* Casoratian, and the recurrence with $c_0(n) = n+1$ gives $C(n) = n!C(0)$ — both matching (4) exactly.

All checks pass. The harness and the Lean sources are listed in Section 9.

7 Epistemic status

- **PROVEN** (Lean, clean cones; Theorem 2): the uniform-in- k step law and product form, the constant-coefficient power form, the integral-domain non-vanishing certificate, and the order-2/order-3 witnesses. No analytic input; the cones are subsets of `{propext, Classical.choice, Quot.sound}` with no `sorryAx`.
- **VERIFIED** (numerically/symbolically; Section 6): the law at $k = 3$ symbolically and for $k = 2, \dots, 7$ by exact-rational checks, plus the named instances.
- **STRUCTURAL** (classical; formalization pending): the *fundamental-system converse* — that when $C(n_0)$ is a unit (over a domain, with c_0 invertible) every solution of (1) is an R -linear combination of the $s^{(t)}$. This is the classical fundamental-set theorem ([1, Ch. 2]); over a field it is immediate from the solution space being pinned by the first k values, as Mathlib’s `LinearRecurrence` already encodes in the constant-coefficient case [6]. We have not yet formalized it in the present uniform-in- k , time-varying setting.
- **CONJECTURED / open**: a concrete simultaneous-Hermite–Padé PCF whose order-3 Casoratian, via Corollary 2, furnishes an Apéry-style irrationality certificate. This is the genuinely open next entry; the present note — together with the converse above — supplies the determinantal machinery it will rely on.

We claim no novelty for Abel’s formula itself, which is classical ([1, Ch. 2],[2],[3]); the contribution is the uniform-in- k machine-checked proof and the opening of the order- ≥ 3 direction in this corpus.

8 Relation to prior work and novelty

Formal-methods landscape. The two standard formal-library treatments of linear recurrences both target the *constant-coefficient* solution theory, not the Casoratian evolution law. Mathlib’s `LinearRecurrence` [6] fixes an order and a tuple of *constant* coefficients `coeffs : Fin order → R`, builds the solution submodule of $\mathbb{N} \rightarrow R$, the `LinearEquiv` sending a solution to its first `order` terms (so solutions are pinned by their initial data), and the characteristic polynomial; it defines no Casoratian and proves no Abel-type evolution law. Eberl’s *Linear Recurrences* entry in the Isabelle Archive of Formal Proofs [9] develops constant-coefficient recurrences via formal power series and polynomial factorisation into an executable closed-form solver, again without the Casoratian/Abel identity. Our object is complementary and, in one direction, strictly more general: the coefficients $c_j(n)$ are arbitrary functions of n (time-varying), the Casoratian is a genuine `Matrix` determinant, and the theorem is the determinantal evolution law uniform in the order k — exactly the non-vanishing / linear-independence certificate (Corollary 2) that Hermite–Padé and Apéry-type constructions consume ([4, 5]). A machine-checked Casoratian does already exist for the order-2 case in another system: the Coq/MathComp formal proof of the irrationality of $\zeta(3)$ [10] defines the 2×2 Wronskian `ba_casoratian` of the two Apéry sequences and derives its first-order (Abel) law, but only for that concrete instance — indeed its source remarks that the order-1 recurrence “should be obtained from something more general.” Having searched Mathlib, the Isabelle AFP, and the Coq/MathComp ecosystem by name (*Casoratian*, *Wronskian*) and by structure, we find no *general-order*, *uniform-in-k* Casoratian Abel formula with time-varying coefficients — precisely the “something more general” the Apéry formalization lacked — which is what the present note provides and machine-checks.

Within this corpus, and a note on hosting. The deposited corpus had formalized the Casoratian only in its order-2 form — the $4/\pi$ PCF Casoratian closed form (Theorem 6.3 of the ladder paper [12]; machine-verified in [11]) and the general order-2 `casoratian_eqG` in `pcf-delta` [13] — so this is the program’s first order- ≥ 3 result and the backbone of a planned Hermite–Padé / Apéry line. We deposit it as a standalone catalogue entry rather than as a Mathlib pull request for two reasons: the result is more general than Mathlib’s constant-coefficient `LinearRecurrence` API and would need new time-varying-coefficient scaffolding to land there, and the corpus requires the core to be hosted and cone-gated inside `pcf-delta`’s `Check.lean`. A Mathlib contribution — a `Casoratian` determinant and its Abel evolution law, specialising to `LinearRecurrence` in the constant-coefficient case — is a natural next step and would convert “not otherwise available” into an externally reviewed library lemma. For the record, the Lean and `sympy` sources label this identity “Abel–Jacobi–Liouville”; the standard discrete name, used here, is Abel’s formula (Elaydi [1]), the triple form naming the continuous-Wronskian analogue (Liouville / Jacobi). The note supplements the SIARC program [14].

9 Code and data availability

The Lean sources are `verify/GeneralCaso.lean` (namespace `PcfGeneralCaso`) and `verify/HigherCaso.lean` (namespace `PcfHigherCaso`); they are built and axiom-cone-checked inside the `pcf-delta` `PcfContinuant` project [13] via `lake env lean Check.lean` (toolchain `leanprover/lean4:v4.30.0`, Mathlib `rev=v4.30.0`). The independent symbolic gate is `harness_caso_k.py` (`sympy` [8]; run `python harness_caso_k.py`), which prints the $k = 3$ symbolic proof, the exact-rational $k = 2, \dots, 7$ verification, and the named instances. Public repository:

<https://github.com/papanokechi/pcf-casoratian-catalogue>.

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